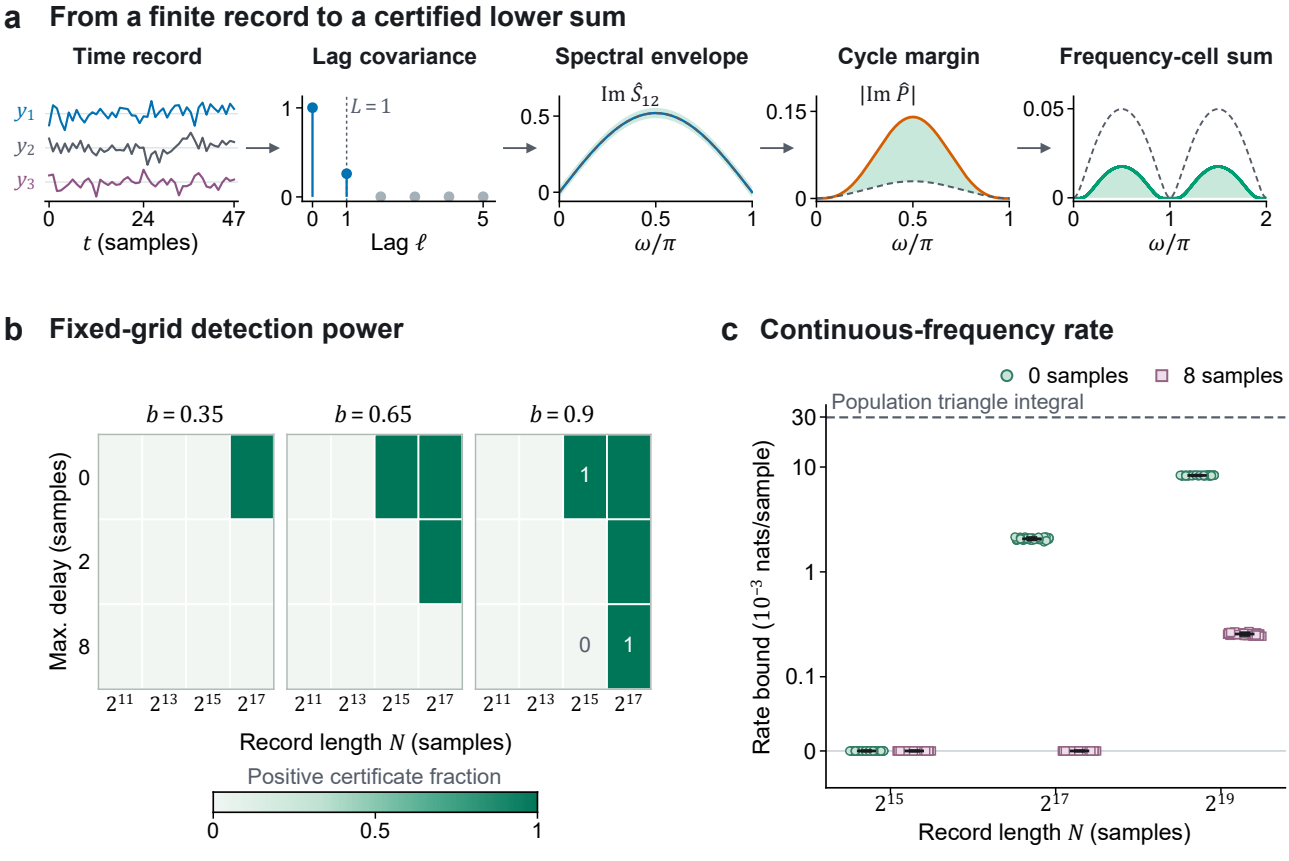

a
From a finite record to a certified lower sum
Time record
Lag covariance
Spectral envelope
Cycle margin
Frequency-cell sum
y_1
y_2
y_3
0
24
47
t (samples)
1
0
$L = 1$
0 1 5
Lag ℓ
0.5
0
$\mathrm{Im}\,\hat{S}_{12}$
0 0.5 1
ω/π
0.15
0
$|\mathrm{Im}\,\hat{P}|$
0 0.5 1
ω/π
0.05
0
0 1 2
ω/π
b
Fixed-grid detection power
$b = 0.35$
$b = 0.65$
$b = 0.9$
Max. delay (samples)
0
2
8
1
0
1
2^{11} 2^{13} 2^{15} 2^{17}
Record length N (samples)
Positive certificate fraction
0
0.5
1
c
Continuous-frequency rate
0 samples
8 samples
Population triangle integral
30
10
1
0.1
0
Rate bound (10^{-3} nats/sample)
2^{15} 2^{17} 2^{19}
Record length N (samples)